

Math 55, Fall 2021 Final Exam

2. True or false:

a. If S is a nonempty subset of the negative rational numbers, then:

$$\exists x \in S (\forall y \in S (x \geq y)).$$

b. If the proposition $p \rightarrow q$ is false then the proposition $\neg p \rightarrow \neg q$ must be true.

c. If A and B are sets such that $A \subseteq B$ then $A \times B \subseteq B \times B$.

d. If A is finite and $f : P(A) \rightarrow B$ is a one to one (i.e., injective) function, then B must be countable. (here, $P(A)$ denotes the power set of A)

e. If B is countable and $f : P(A) \rightarrow B$ is a one to one (i.e., injective) function, then A must be finite. (here, $P(A)$ denotes the power set of A)

f. The set of all finite connected simple graphs with vertex sets $\{1, \dots, n\}$ for $n \geq 1$ is countable.

g. If $\gcd(a^2, b^3) > 1$ then a and b cannot be relatively prime.

h. Suppose I draw one card uniformly at random from a deck of 52 cards. Then the events $E = \{\text{the card is a spade}\}$ and $F = \{\text{the card is an ace}\}$ are independent.

i. The number of even degree vertices in a simple graph must be even.

j. For every graph G on n vertices, either G or its complement with respect to K_n must be Eulerian (i.e., contain an Euler circuit). (clarification: *The complement $\overline{G}$ of $G = (V, E)$ with respect to K_n is the graph $\overline{G} = (V, \{\{u, v\} : \{u, v\} \text{ is not an element of } E\})$.)*

3. Find an integer x satisfying the equation:

$$4x^{21} \equiv 2 \pmod{5},$$

or show that no such integer exists.

4. Find an integer x satisfying the two congruences:

$$x \equiv 1 \pmod{3}$$

$$2x \equiv 0 \pmod{6},$$

or show that no such integer exists.

5. Suppose n students are each independently assigned a birthday in $\{1, \dots, 365\}$, uniformly at random. Let T_n be the random variable equal to the number of unordered triples (i.e., sets of three distinct students) of students with the same birthday.

- (a) Show that if $n \geq 120$ then $\mathbb{E}T_n \geq 1$.
- (b) How large must n be in order to guarantee that there are three students who share the same birthday, i.e.

$$\mathbb{P}(T_n \geq 1) = 1?$$

Find the least such n .

6. Suppose I flip n independent biased coins such that the j th coin has probability j/n of being heads, for $j = 1, \dots, n$. Let H_n be a random variable equal to the total number of heads.
 - (a) What is the expectation of H_n ?
 - (b) What is the variance of H_n ?
 - (c) Use Markov's inequality to derive an upper bound on $\mathbb{P}(H_n \geq 9n/10)$.
7. Show that if a simple graph G has n vertices and at least n edges, then G must contain a cycle.
8. In this question all graphs are simple and n is an arbitrary positive integer.
 - 8.1. Show that every tree on n vertices has exactly two 2-colorings.
 - 8.2. (a) Recall that the path graph on n vertices has vertex set $V = \{1, \dots, n\}$ and edge set $E = \{\{i, i+1\} : i = 1, \dots, n-1\}$. How many 3-colorings does the path graph on n vertices have? (hint: use a recurrence relation).
 - (b) Let k be a positive integer. Show that every tree on n vertices has exactly the same number of k -colorings. What is this number of colorings? (hint: consider your answers to previous parts of this question)
9. Let n be an even positive integer.
 - (a) Show that if the degree of every vertex in a simple graph G on n vertices is at least (i.e., greater than or equal to) $n/2$, then G must be connected.
 - (b) Give an example of a disconnected simple graph on n vertices in which every vertex has degree $(n/2) - 1$.